

Technical Document #298: Retrieval Augmented Generation - Comprehensive Overview

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Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Chunking splits documents into smaller segments for embedding.
- Context window optimization selects the most relevant information to fit within the LLM's context limit.
- Semantic search finds documents based on meaning rather than exact keywords.